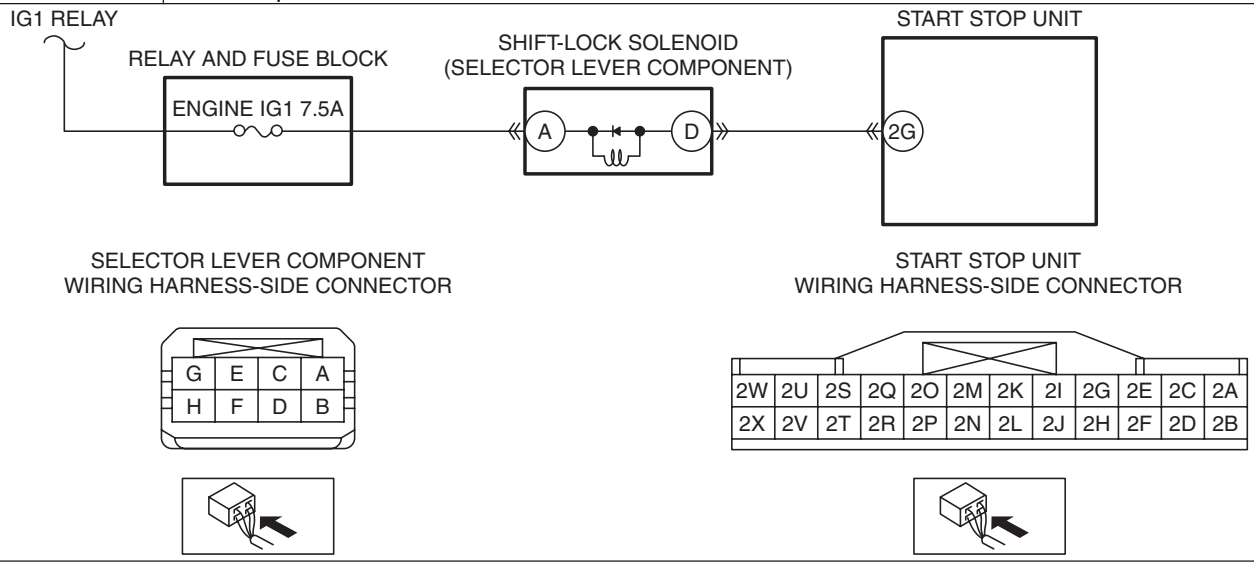
## DTC P0930:71 [START STOP UNIT]

id0902p6023400

| Description | Shift-lock solenoid circuit malfunction |
|---|---|
| **Detection condition** | • Start stop unit detects the shift-lock solenoid is stuck on for **30 min or more** with the ignition switched ON (engine off or on). |
| **Fail-safe** | • Inhibits the shift lock control. |
| **Possible cause** | • Selector lever component connector or terminal malfunction<br>• Shift-lock solenoid power supply circuit malfunction<br>— Short to ground in wiring harness between ENGINE IG1 7.5 A fuse and selector lever component terminal A<br>— ENGINE IG1 7.5 A fuse malfunction<br>— Open circuit in wiring harness between IG1 relay and selector lever component terminal A<br>• Start stop unit connector or terminal malfunction<br>• Short to ground in the wiring harness between selector lever component terminal D and start stop unit terminal 2G<br>• Open circuit in the wiring harness between selector lever component terminal D and start stop unit terminal 2G<br>• Shift-lock solenoid malfunction<br>• Start stop unit malfunction |




### Diagnostic Procedure

| Step | Inspection | | Action |
|---|---|---|---|
| 1 | **INSPECT SELECTOR LEVER COMPONENT CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Disconnect the selector lever component connector.<br>• Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection.<br>• Is the connector normal? | Yes | Go to the next step. |
| | | No | Repair or replace the connector, then go to Step 7. |

| Step | Inspection | | Action |
|---|---|---|---|
| 2 | **INSPECT SHIFT-LOCK SOLENOID POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT OR SHORT TO GROUND**<br>• Verify that the selector lever component connector is disconnected.<br>• Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Switch the ignition ON (engine off or on).<br>• Measure the voltage at the selector lever component terminal A (wiring harness-side).<br>• Is the voltage **B+**? | Yes | Go to the next step. |
| | | No | Inspect the ENGINE IG1 7.5 A fuse.<br>• If the fuse is blown:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between ENGINE IG1 7.5 A fuse and selector lever component terminal A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>• Replace the fuse.<br>• If the fuse is damaged:<br>— Replace the fuse.<br>• If the fuse is normal:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between IG1 relay and selector lever component terminal A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 7. |
| 3 | **INSPECT START STOP UNIT CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Disconnect the start stop unit connector.<br>• Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection.<br>• Is the connector normal? | Yes | Go to the next step. |
| | | No | Repair or replace the connector, then go to Step 7. |
| 4 | **INSPECT SHIFT-LOCK SOLENOID CIRCUIT FOR SHORT TO GROUND**<br>• Verify that the selector lever component and start stop unit connectors are disconnected.<br>• Inspect for continuity between selector lever component terminal D (wiring harness-side) and body ground.<br>• Is there continuity? | Yes | Refer to the wiring diagram and verify whether or not there is a common connector between selector lever component terminal D and start stop unit terminal 2G.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>Go to Step 7. |
| | | No | Go to the next step. |

| Step | Inspection | | Action |
|---|---|---|---|
| 5 | **INSPECT SHIFT-LOCK SOLENOID CIRCUIT FOR OPEN CIRCUIT**<br>• Verify that the selector lever component and start stop unit connectors are disconnected.<br>• Inspect for continuity between selector lever component terminal D (wiring harness-side) and start stop unit terminal 2G (wiring harness-side).<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between selector lever component terminal D and start stop unit terminal 2G.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 7. |
| 6 | **INSPECT SHIFT-LOCK SOLENOID**<br>• Inspect the shift-lock solenoid. (See SHIFT-LOCK SYSTEM INSPECTION.)<br>• Is the shift-lock solenoid normal? | Yes | Go to the next step. |
| | | No | Replace the selector lever component, then go to the next step.<br>(See AUTOMATIC TRANSAXLE SHIFT MECHANISM REMOVAL/INSTALLATION.) |
| 7 | **VERIFY THAT REPAIRS HAVE BEEN COMPLETED**<br>• Always reconnect all disconnected connectors.<br>• Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].)<br>• Switch the ignition ON (engine off or on) and wait for **30 min or more**.<br>• Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].)<br>• Is the same DTC displayed? | Yes | Repeat the inspection from Step 1.<br>• If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.)<br>Go to the next step. |
| | | No | Go to the next step. |
| 8 | **VERIFY IF OTHER DTCs DISPLAYED**<br>• Are any other DTCs displayed? | Yes | Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [START STOP UNIT].) |
| | | No | DTC troubleshooting completed. |